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Effect of dust on the transparent cover of solar collectors

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Abstract

Research and development in photovoltaic (PV) systems has usually been concentrated in studies on radiation availability, efficient operating strategies, design and sizing of these systems. On the other hand, the influence of dust on the performance of PV systems has not been given much attention. The work described here contributes considerably to overcome this deficit. To validate that concept, we have developed an experimental set up involving 100 glass samples with different tilt (β) and azimuth angles (γ). The transmittance of the glass was evaluated at regular intervals over a period of about seven months and after every thunderstorm in the surrounding area.

Although it is only a short term view, the preliminary results indicate that the reduction in glass normal transmittance depends strongly on the dust deposition density in conjunction with plate tilt angle, as well as on the orientation of the surface with respect to the dominant wind direction. With this consideration, one sees that as the dust deposition density goes from 15.84 g/m^2 (for glass sample installed at a tilt angle of 0°) to 4.48 g/m^2 (for glass sample installed at a tilt angle of 90° and oriented with 135° deviation from north), the corresponding transmittance diminishes by approximately 52.54–12.38%, respectively. The evolutions of the output power variation with increasing cell pollution were also examined. It has been found that the slope of the best straight line passing through the data points of the solar cell installed at a 45° angle facing south suggests a decrease in the output power of about 17.4% per month.
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Keywords: Solar cells; Dust deposition density; Gradual reduction of transmittance; Multi-position test facility; Solar radiation; Outdoor evaluation; Incidence angle

1. Introduction

The performance of a solar energy system (whether using thermal or PV collector) is influenced by the ability of the cover glass to transmit solar radiation to the collection surface besides other factors, such as the

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intensity of solar radiation, the tilt angle of the absorbing surface, the properties of the materials etc. One of the factors mentioned is the gradual degradation of transmittance that depends on the accumulation of dust. In reading the PV literature, one sees little concern for the problem of dust. The limited published works indicate, however, that the problem has not even been well investigated. For example, Salim et al. [1] constructed a PV test system at a solar village near Riyadh, Saudi Arabia, to study the effect of long term accumulation of dust on the PV array energy output. With the fixed tilted array at 24.6°, the monthly energy reduction for the unclean array was obtained by comparing its performance with an identical array, which was cleaned daily. The reduction in the energy output from the unclean array reached 32% at the end of eight months. This study, however, did not present information about the dust level in the test site or its physical properties. Moreover, the authors' results are totally misleading because in highly polluted sites, these reductions in the energy would result only after a few weeks, not eight months. For instance, work conducted by Hassan et al. [2] shows that the degradation progresses rapidly during the first 30 days of exposure. The results imply that the depression of the efficiency is 33.5% after one month and increases to 65.8% after six months without panel cleaning.

In similar investigations, Garg [3] has measured the normal transmittance of direct radiation through glass and found that over a period of 30 days, the transmittance decreased from 90% to 30% for a horizontal mounting. Similar measurements were made in Kuwait by Sayigh et al. [4], who observed 64%, 48%, 38%, 30% and 17% reduction in the transmittance of the glass plates after 38 days of exposure to the environment with tilt angles of 0°, 15°, 30°, 45° and 60°, respectively. Here again, interpretation is difficult since the dirty and clean tests were done on different days, and it does not appear that insolation measurements were made to reduce the data to a common base. Probably the most complete study is that by El-Shobokshy and Hussein [5]. In the laboratory, they polluted PV surfaces with different kinds of dust and measured the electric output of the cells under different conditions. It may be concluded from this study that the effect of dust on the performance of PV cells can no longer be correlated to the exposure time in a given site as suggested by many researchers. The results also indicate that the nature of the dust, such as dust material, its size distribution and the dust deposition density, has a strong influence on the loss of output power from the PV panels. This study, however, contains some restrictions. Probably the most important of these is that all the PV surfaces they prepared were polluted under zero wind conditions since, in natural circumstances, low winds significantly affect the sedimentological structure of dust coatings on flat surfaces [6,7]. Another restriction of El-Shobokshy and Hussein's work is that no natural desert dust was used in their experiments.

On the other hand, there are a number of studies in the literature concerning the effect of dust on solar devices in rainy regions. In one of the earlier studies, Hottel and Woertz [8] conducted 3 month performance tests on collectors having a 30° tilt angle and located in an industrial district near a power plant and 92 m away from a four track railroad. Experimentally, they found that the effect of dirt on the collector's net performance was surprisingly small (4.7%), while the calculations indicated 2.7% maximum reduction in the glass transmittance. They attributed these small values to the collector's self-cleaning action, incorporated due to snow and rainfall in Boston, USA. In a similar approach, but for tilt angles between 0° and 50°, the data of Dietz [9] showed a reduction in the solar radiation reaching the collector by a factor of 5% due to dirt accumulation. This finding was recently affirmed by Michalsky et al. [10], when two pyranometers were set side by side for two months in Albany, New York. One of the pyranometers was cleaned daily, while the other was left unattended, but during this period, the test site had some rain at least once every 10 days. Even so, less than 1% decrease in the values from the unclean pyranometer was observed. The above literature survey indicates that the effect of the accumulation of dust on the transparent covers of solar devices has not received much attention, especially in countries with dusty climates. On the other hand, the limited published studies do not provide enough information about the dust deposition rate in the test site nor its physical properties. Thus, more data are required for design purposes and to generalize the reduction in the performance of solar cells due to accumulation of dust.

This article presents the results of a study done to assess the influence of natural soil dust on the transmittance of a glass cover under arid climatic conditions. Attention will also be paid to the chemical and mineralogical composition of the dust coating on the PV's transparent cover and to the effect it exerts on cell performance.

2. Basic specification of the experiment

Here, we provide the groundwork for discussing the influence of dust on the transmittance of 3 mm thick glass used commonly in solar collectors. 100 mm × 100 mm size samples of glass were cut and installed over a wooden frame in eight different orientations (north, northeast, east, southeast, south, southwest, west and northwest) with seven tilt angles (0°, 15°, 30°, 45°, 60°, 75° and 90°) and exposed to the environment. Care is taken to ensure that accumulations of dust and other dirt are not removed by any agents other than wind and rain. Fig. 1 shows a photograph of the outside view of the multi-position test facilities. The cleaning operations were conducted monthly, and one set of samples was cleaned after approximately seven months. By this way, a detailed study has been conducted to quantify the reduction in transmittance of the glass samples. We also investigate the reduction in the cells' output power due to dust accumulation. The PV cells used in the experiment were monocrystalline silicon solar cells (20 mm × 20 mm).

Seven solar cell samples were fixed facing south at 0°, 15°, 30°, 45°, 60°, 75° and 90° tilt from the horizontal and six samples were kept at 30° and 90° but facing east, west and north. The cell characteristics were measured indoors using the solar simulator. All cells showed a uniform performance as their short circuit currents, open circuit voltages and I–V curves were close to each other.

A weekly measurement of the cell parameters has been taken for four values of relative atmospheric air masses (AM4; AM3; AM2 and AM1.7). We selected these particular air masses because, at the latitude of the station, the sun reaches the elevation of these values in a large part of the year. Thus, with this selection, it is possible to study the diurnal and seasonal variation of the performance of the PV cells. In order to minimize the variation in temperature and solar radiation, the data has been recorded during clear and sunshine days with a wind speed of about 2 m/s.

2.1. Brief description of the apparatus

According to the aforementioned survey, the performance of a solar collector drops progressively as dust is accumulated on its surface. The rate of decrease in the performance depends mainly on the rate of dust deposition. To measure the quantity of dust settling onto the glass surfaces, the glass samples were weighed under controlled conditions before the experiment using a Mettler PJ3000 balance with an accuracy of 0.001 g. The samples with settled dust were weighed again, and the weight of the dust particles accumulated was calculated from the increase in mass. The weight was divided by the surface area from which they were collected to give the dust deposition density in g/m^2. The transmittance of the glass samples were measured using a small apparatus described by Fig. 2. Mineralogical analyses of the natural soil dust collected onto the glass surfaces have been performed by X-ray diffraction. The methodology of X-ray diffraction is described by Ignasi et al. [11].



Fig. 1. Wooden frames covered by glass samples exposed to the environment.

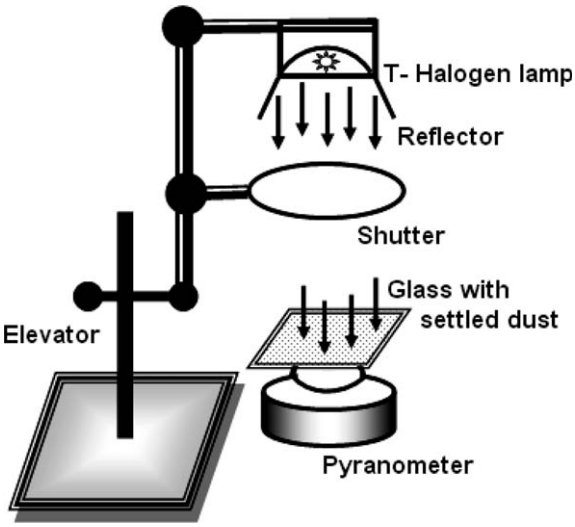


Fig. 2. Schematic diagram of the apparatus used for measuring the glass normal transmittance.

3. Location and environment

The PV system and measuring apparatus were installed on the terrace of the new building of the National Research Institute of Astronomy and Geophysics (Lat. 29°52′N and Long 31°20′E), where it is located at about one and one half kilometers northeast of the town on a sandy limestone plateau 120 m above sea level. The plateau is one of a series whose heights gradually increase towards the east. The topography around the site is rugged, particularly in the north and in the northeast. Unfortunately, a strong pollution is produced by the establishment of various industries in the neighborhood. The local industry includes four cement factories located from the north in Tura to the south in Tebeen, engineering industries (e.g. an automobile factory, a pipe and tube factory) and iron and steel works. There is also a main railway line at a distance of only 1.5 km. Pollution by iron dust from railway lines seems to favor development of permanent pollution. The prevailing wind direction is from the north and northeast, which represents about 50% of the total directions. This leads to an important result, that is, cement factories contribute approximately 50% to the air pollution of the site. Fig. 3 demonstrates this fact and shows the close match between pure cement and dust settling onto the surface of the flat plate at the considered site [12].

The climate in the considered site is usually affected by the Asiatic low in summer, the Sudan low in spring and autumn and the Cyprus low in winter. Because of the Asiatic low, the vapor pressure increases in the summer months and stratus forms of clouds are formed. The migration of the Sudan low actuates the Khamasin depressions, which are maintained by continental depressions over the western desert of the country. The weather associated with these depressions is generally hot, dry and dusty. It should be noted that the composition of the dust deposited on the transparent covers during a Khamasin storm was similar to that deposited during normal weather conditions, indicating that the composition of the dust deposited during the storm over a wide area of the desert around Helwan was uniform.

Cyprus lows have marked influences on atmospheric conditions; violent wind, rain, showers and dust storms inland frequently happen due to them [13]. The average summer temperature is 27.5 °C, while that of winter is 15.4 °C with an average difference between maximum and minimum temperature of 12.6 °C. This great temperature difference promotes the formation of dew at dawn as the relative humidity of the air becomes generally high. Dew formation aids dust settling on flat surfaces, while evaporation, on the other hand, reinforces dust adhesion to these surfaces. Relative humidity fluctuates between 41% in May and 62% in January, with a visibility of about 5 km. This restricted visibility is the result of the presence of solid particles in the atmosphere.

The prevailing wind velocity with an average of 3.7 m/s but gusts up to 5–6 m/s may be experienced in early morning and late afternoon. Dust storms also occur during April and May when the Khamasin winds blow

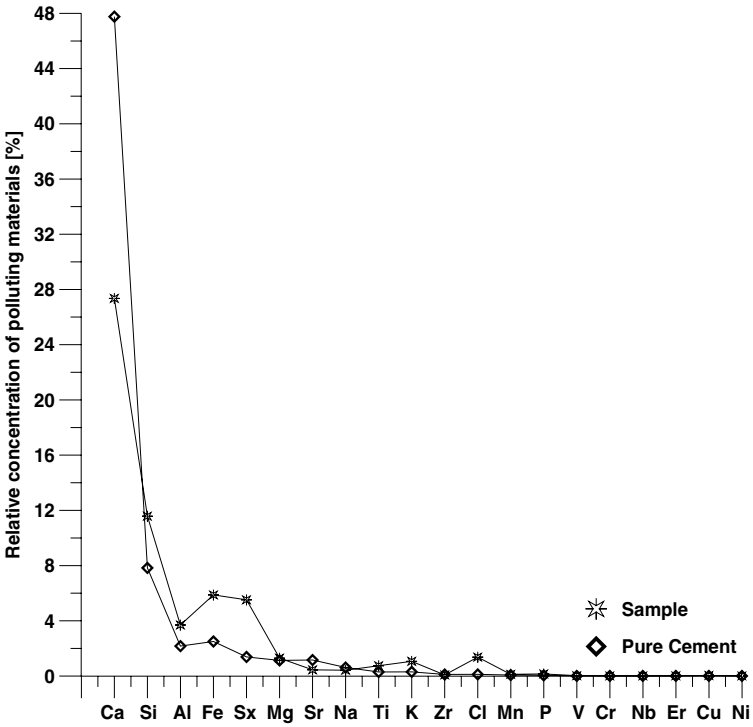


Fig. 3. Comparison between natural soil dust settling onto the flat plate and pure cement.

over the Egyptian western desert with a wind speed of the order of 10 m/s or more. The site of the experiment has approximately 18 mm of rainfall per year. This relatively small amount of precipitation occurs almost exclusively during the months of November through April, mostly in the form of scattered showers. These showers usually last for a few minutes at a time and only rarely for an hour or more. Usually, no rain occurs during the remaining six months. In addition to the aforementioned variables, monthly values of the sunshine duration can be as high as 358 h in June with the least values (197 h) in December.

4. Chemical composition of the polluting material

The question of the trace power of regional elemental characteristics has been treated in a number of publications. As summarized in some reference papers [14–17] single element traces or ratios of elemental concentrations can be used for studying the nature of the main emission sources in the region as well as for pinpointing the source areas of transported pollutants to the site of observation. Among others, sulfur is an indicator for fossil fuel burning, fine fraction vanadium and nickel are excellent indicators for residual fuel burning, copper and zinc are often observed to be linked with traffic (together with lead), arsenic is a tracer for coal combustion, selenium has a variety of sources but mostly fossil fuel burning, leaded gasoline; smelters and waste incinerators are all possible sources of lead [18].

According to elemental and mineralogical analysis of the dust accumulated on the transparent covers of solar cells, the dust particles were composed mostly of quartz and calcite with smaller amounts of dolomite and clay minerals. The major constituents were Si and Ca; minor constituents were Al; Fe; Mg; K and Na. The characteristic features of the dataset presented in Fig. 4 can be summarized as follows: silicon, Si, and calcium, Ca, are regarded as the important components of PM_{10} measured in the considered site. The presence of coarse Si can be ascribed to erosion from the desert land surrounding the monitoring site and also by the effects of human activities resulting in disturbances to the soil. Ca, can be ascribed partly to emanation from industries such as cement plants. Ongoing building activities and degradation of building elements can be considered as additional sources for increasing calcium level. Sulfur, S, indicates the determining contribution of

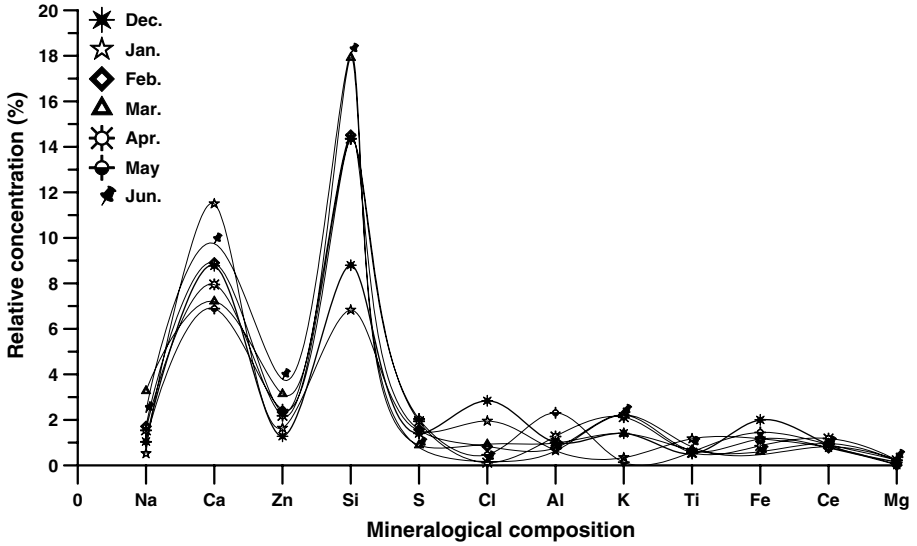


Fig. 4. X-ray analysis of the polluting material showing the most important organic components.

long range transport processes during which gaseous sulfur emission is subject to a continuous conversion to particulate secondary sulfur. The increased values in sulfur, S, can be considered as an indication for the possible influence of Shoubra El-Kheima (a highly industrialized area) on Helwan air. The source regions of Shoubra El-Kheima do not contain a single large source of air pollutants but a rather large number of smaller sources.

These sources include industrial boilers; some smelting activity; paper mill operations and electrical power plants. Potassium was ascribed to mixed natural (i.e. soil derived contribution as in the silicon case) and anthropogenic sources. Low chlorine, Cl and sodium Na concentrations can be understood as a strong sea derived contribution. Small concentrations of aluminum Al, zinc Zn, manganese Mg, titanium Ti and iron Fe indicate the importance of soil derived dust of traffic origin.

5. Influence of dust deposition on the transmittance and on energy yield

5.1. The influence of the orientation and tilt angle on transmittance

The investigation of dust accumulation on the glass samples in a natural condition was conducted outdoors during the months of December 2004 to June 2005. The resulting monthly average of the total suspended particles deposited on the glass surfaces is shown in Fig. 5. It appears from the information in Fig. 5 that as tilt angle increases, dust particles tend to roll from the surface.

The quantity of dust accumulated on the glass samples installed at different orientations with the same tilt was approximately equal. Nonetheless, it was 8.03 g/m^2 for the glass sample having a tilt angle of 15° and oriented with a 45° deviation from north (i.e. 15° NE). A possible reason for this is that the northeast winds could affect the atmosphere by fine particles of various origins, mainly from the emission sources of cement factories.

In spite of its significance, the increases of accumulated dust may also alter the performance of the solar cells by diminishing the transmittance of the transparent cover and thereby influencing the amount of energy yield. Therefore, the average reduction in transmittance for various dust deposition densities were calculated and are plotted in Fig. 6. The reduction in transmittance is given as the difference of transmittance observed between a clean sample and a sample with dust. It is evident from Fig. 6 that the reduction in transmittance is directly caused by the accumulation of dust on the glass surface. In fact, as the dust deposition increases, the transmittance reduction increases too but with a progressively decreasing rate until reaching its upper limit. Thereafter, the effect of dust deposition vanishes. Accordingly, it is reasonable to seek the following simple

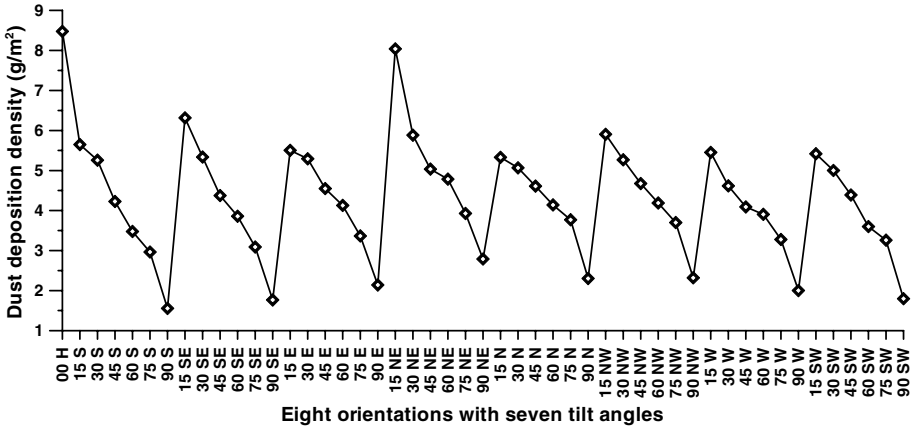


Fig. 5. Quantity of dust accumulated on glass samples installed in eight different orientations with seven tilted angles in an arid location.

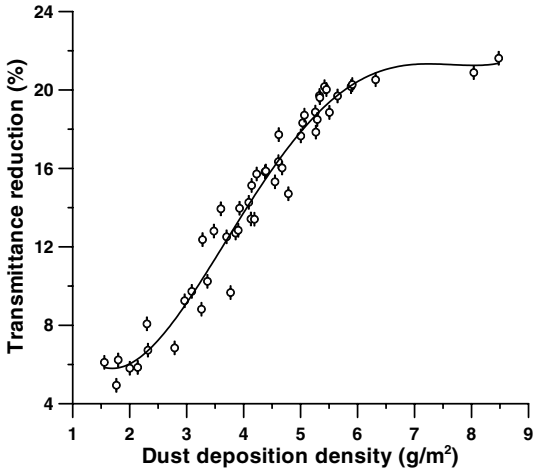


Fig. 6. Reduction in transmittance as a function of dust deposition density.

correlation form to reflect this natural behavior between dust deposition density ρ_D (g/m^2) and reduction in transmittance $\Delta\tau$ (%).The comparison between the correlation predictions and the measured data is shown by the solid line in Fig. 6. The frequency diagram of the difference between the synthetic and observed values is shown in Fig. 7. The correlation coefficient (R), F-test, mean bias error (MBE) and root mean square error (RMSE) are also presented in the figure. Generally, these results are considered as an adequate accuracy for such predictions

$$\Delta\tau = 0.0381\rho_D^4 - 0.8626\rho_D^3 + 6.4143\rho_D^2 - 15.051\rho_D + 16.769 \quad (1)$$

If care is taken in considering the corresponding regional climatic differences as well as the plate mounting, the results of this correlation form can be generalized and transferred to other sites.

To quantify the influence of tilt on the reduction in transmittance, the reduction in transmittance for various tilt angles were calculated and are listed in Table 1. Table 1 shows the degradation in the glass normal transmittance due to dust accumulation in terms of eight different orientations with seven tilt angles. From Table 1, it is inferred that the reduction in transmittance was maximum (27.62%) for the glass sample having a tilt of 0° because the accumulation of dust (8.48 g/m^2) was more on this as compared to the samples having a tilt of 15° (≅20.18%), 30° (≅18.47%), 45° (≅15.96%), 60° (≅13.62%), 75° (≅10.82%) and 90° (≅6.32%), whereas the reduction in transmittance was minimum (4.94%) for the glass sample installed at a tilt angle

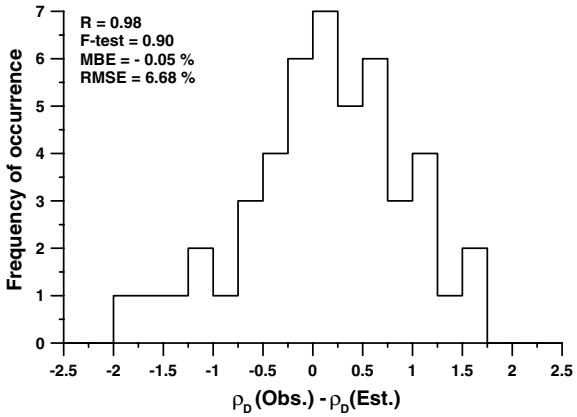


Fig. 7. Frequency distribution of the difference between observed and modeled values.

Table 1
Reduction in normal transmittance as a function of various directions and tilt angles

Tilt angle	Orientation of the glass samples							
	N	NE	E	SE	S	SW	W	NW
15°	19.71%	**20.89%**	20.52%	20.17%	20.28%	20.17%	20.02%	19.69%
30°	18.72%	**18.86%**	18.50%	19.61%	18.87%	17.66%	17.72%	17.85%
45°	16.34%	**18.32%**	15.32%	15.82%	15.72%	15.86%	14.27%	16.02%
60°	15.13%	14.71%	13.42%	12.71%	12.81%	13.94%	12.85%	13.41%
75°	9.67%	**13.97%**	10.24%	9.73%	9.25%	8.82%	12.37%	12.51%
90°	8.07%	6.84%	5.85%	**4.94%**	6.11%	6.23%	5.81%	6.73%

Average reduction in transmittance for glass sample having a tilt angle of 0° is 27.62%.

of 90° and oriented with 135° deviation from north (i.e. 90° SE). For northeast facing, where cement plants were located, the reduction in transmittance was maximum for glass samples having a tilt of 15° (20.89%), 30° (18.86%), 45° (18.32%) and 75° (13.97%), respectively.

In addition, a closer examination of the glass surfaces of one set of samples that were never cleaned until 30 June 2005 revealed that a visible but transparent layer of dissolved and subsequently re-crystallized salts had accumulated on the surface (see Fig. 8). These layers, which were by this time too thick to be dislodged by rain showers contributed significantly to the optical measurements. The reduction in transmittance of these samples were measured and are listed in Table 2.

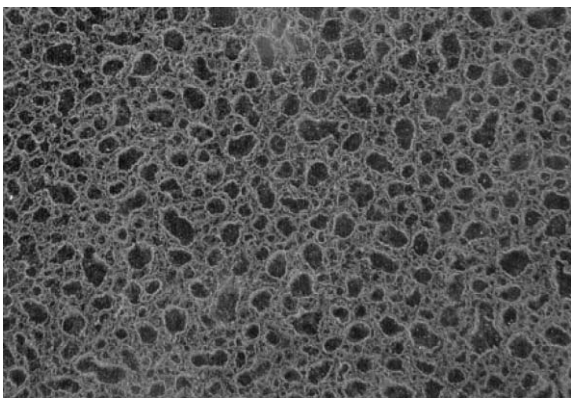


Fig. 8. Build up of mineral deposits is visible on the surface of the glass sample.

Table 2
Reduction in transmittance ($\Delta\tau$ in %) as a function of dust deposition density (ρ_D in g/m^2) for glass samples installed at various directions and tilt angles

Tilt angle	Orientation of the glass samples															
	N		NE		E		SE		S		SW		W		NW	
	$\Delta\tau$	ρ_D	$\Delta\tau$	ρ_D	$\Delta\tau$	ρ_D	$\Delta\tau$	ρ_D	$\Delta\tau$	ρ_D	$\Delta\tau$	ρ_D	$\Delta\tau$	ρ_D	$\Delta\tau$	ρ_D
15°	41	12	46	11	45	12	47	13	46	14	46	14	43	12	44	12
30°	40	11	45	13	46	13	47	11	46	13	44	12	41	11	47	12
45°	40	11	41	12	42	12	44	12	45	13	43	14	42	12	44	10
60°	36	10	38	9	42	11	39	11	41	12	39	12	39	11	38	9
75°	27	7	26	6	29	8	27	8	29	9	41	7	28	7	25	6
90°	15	3	15	3	15	4	12	2.5	13	5	15	3	13	3	17	4

Average reduction in transmittance for glass sample having a tilt of 0° is 52.54% ($\rho_D = 15.84$ g/m^2).

In light of this result, one sees that as the dust deposition density goes from 15.84 g/m^2 (for glass sample installed at a tilt angle of 0°) to 2.48 g/m^2 (for glass sample installed at a tilt angle of 90° and oriented with 135° deviation from north), the corresponding transmittance diminishes by approximately 52.54–12.38%, respectively. Obviously, large transmittance reductions are encountered as the tilt angle is decreased, which is directly attributed to the increase in dust deposition on the plate's surface. The biggest amount of dust is attained at the tilt angle $\beta = 0°$. As β increases, the dust density starts to decrease, but gradually, until it levels off at a tilt of 90°.

The total reduction in the transmittance samples that were not cleaned until 30 June 05 was very large. However, the performance of the PV cells would probably not be affected to the same extent as the transmittance, because the absorber plate would be expected to capture the forward scattered solar radiation, which would not be available to the normal incidence.

5.2. The influence of dust deposition on the PV energy yield

The purpose of this section is to gain some quantitative understanding of the rate at which desert dust causes degradation of the output of unattended PV modules. Unlike calculations using synthetic data or theoretical considerations, realistic experience of dust influences can be obtained from Figs. 9 and 10. As expected, the extent to which surface dirt accumulates on the solar cells' glass covers and thereby reduces its output was found to be a function of the quantity of the settled dust in conjunction with plate tilt angle, as well as on the orientation of the surface with respect to the dominant wind direction. For the day dated 8 January, the area received 3 mm of rain, which has resulted in a wash out of the dust deposited onto the PV cell's transparent cover. Subsequently, the transmittance increased and the first power peak was observed. Similarly, there was rainfall on 8 March and 9 March 2005 ($\cong$11 mm) and again the transmittance was increased. Therefore, the power yields from dusty cells revealed it to be indistinguishable from a clean one, hence a second power peak was observed. There was no rainfall after 10 March 2005 to the end of the experiment. Hence; there was a large decrease in the transmittance, thereby influencing the amount of energy yield. Thus, the sparse desert rainfall at the considered site sufficed to restore the photovoltaic solar cells to their initial condition.

Figs. 9 and 10 indicate that the PV cell installed at a 45° angle facing south generates higher power regardless of the seasonal difference. The slope of the best straight line passing through the data points of this cell suggests a decrease in the output power of about 17.4% per month. It also shows that the PV cell installed at a 90° angle facing north generates the least power, and that the output of the PV cell installed at a 30° angle facing north largely varies by the season. The main factor that influenced the power of the PV cell installed at 30° facing north is considered to be the effects of the dirt settled on the surface of that cell. Other possible factors that influenced the power yield are the temperature increase, spectrum and the reflection due to the solar incidence angle.

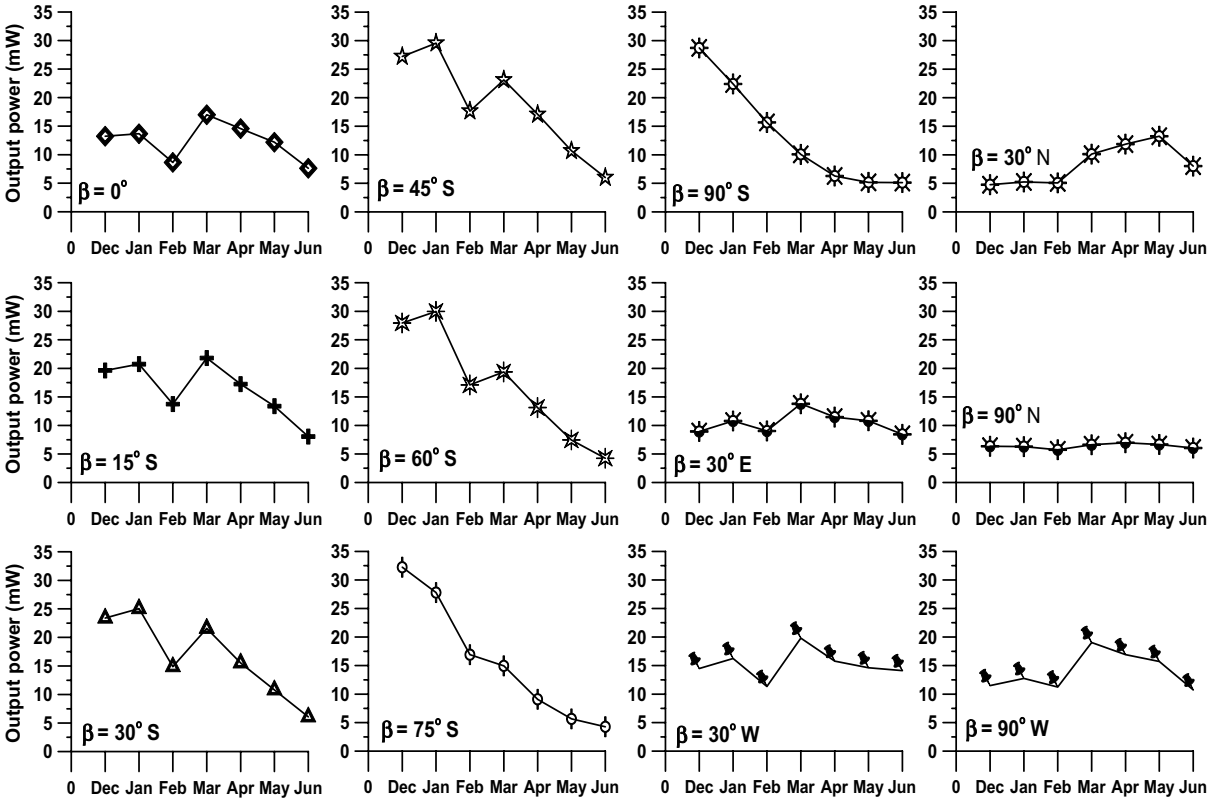


Fig. 9. Effect of the natural soil dust on the output power extracted from PV cells with different installation patterns.

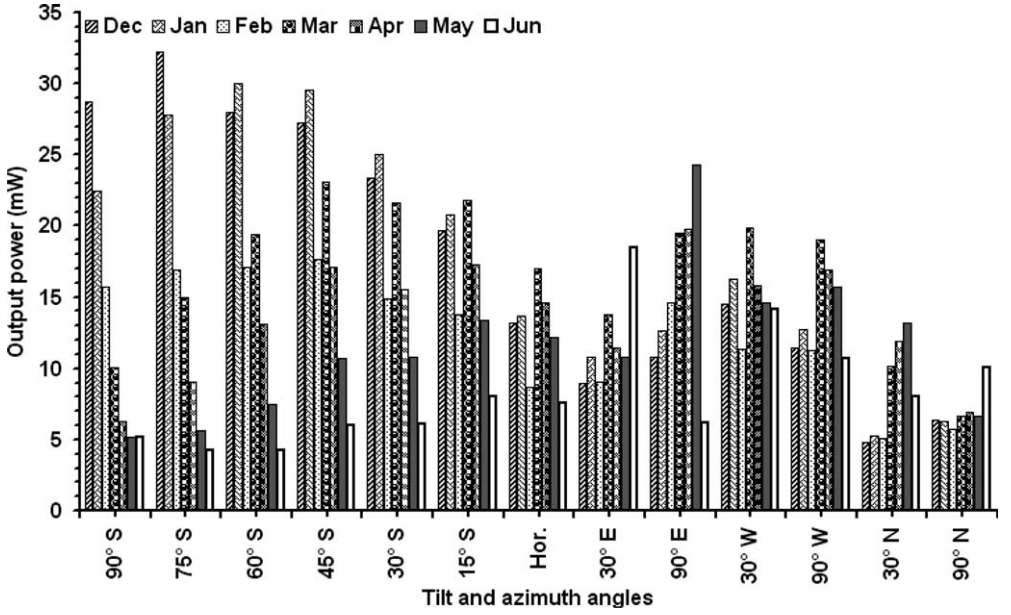


Fig. 10. The PV output power as a function of various directions and tilt angles.

6. Conclusion and recommendations

PV modules are often the power supply of choice for communication systems and other rarely visited installations, particularly in locations where access is difficult. In the case of desert sites, there is concern that the continuous accumulation of dust on the modules might eventually neutralize their effectiveness. The purpose of this study is, accordingly, to gain some quantitative understanding of the rate at which desert dust causes degradation of the output of unattended PV modules. The reduction in transmittance due to various dust deposition densities has been measured and a correlation relating the dust deposition density with the normal transmittance was given.

The results showed that the reduction in glass normal transmittance depends strongly on the dust deposition density in conjunction with tilt angle, as well as on the orientation of the surface with respect to the dominant wind direction. With this consideration, one sees that as the dust deposition density goes from 15.84 to 4.48 g/m^2, the corresponding transmittance diminishes by 52.54–12.38%, respectively. Obviously, large transmittance reductions are encountered as the tilt angle is decreased, which is directly attributed to the increase in dust deposition on the plate.

Further, there is a significant degradation in glass transmittance for the glass sample having a tilt angle of 15° and oriented with a 45° deviation from north. A possible reason for this is that the northeast winds could affect the atmosphere by fine particles of various origins, mainly from the emission sources of cement factories. The following simple correlation form can be used to reflect the relationship between dust deposition density and the reduction in transmittance, $\Delta\tau = 0.0381\rho_D^4 - 0.8626\rho_D^3 + 6.4143\rho_D^2 - 15.051\rho_D + 16.769$. If care is taken in considering the corresponding regional climatic differences as well as the plate mounting, the results of this correlation form can be generalized and transferred to other locations. Here also, a substantial reduction in output power took place as a result of dust accumulation. It is clear that the reduction in output power is erratic and in zigzag manner. This was due to complex phenomena during the test period. There was rainfall, high winds and sand storms. Furthermore, large daily variations in humidity frequently occurred during the considered period, with a minimum of about 41% and a maximum of over 62%. Thus, some dew is assumed to have formed on the glass covers, and this tends to coagulate the dust. Therefore, for moderately dusty places, weekly cleaning of the glass covers is strictly recommended as part of the maintenance routine, but the equipment should be cleaned immediately after a dust storm to retain its nominal operating efficiency.

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